

Protein and Amino acid Metabolism

Biological importance

- Body proteins have life times. They undergo degradation and re-synthesis. about 6 gm of protein is synthesized and broken down per kg body weight per day.
- Aged proteins, damaged or modified proteins and non-functional proteins of the body undergo degradation(i.e enzyme control)-quality control mechanism.
- Protein degradation play an important role in shaping tissues and organs during pregnancy and development.
- Protein synthesis and degradation is an integral part of cellular adaptation to changed environment.
- Amino acids are needed for the formation of specialized products like: **hormones, purines, pyrimidines, porphyrins, vitamins, amines, creatine and glutathione.**

Protein turnover

- Like intermediates of metabolic pathways body proteins are in dynamic state.
- Continuous synthesis and degradation of protein is known as **protein turnover**.

Note: The rates of protein synthesis and degradation vary according to physiological needs.

- The rate of protein synthesis: high during growth, lactation and post operative recovery.
- The rate of protein degradation: high In starvation, cancer, fever and during morphogenesis.

Protein half life

- Body proteins have life times which are expressed in terms of half-life.
- Half life of proteins ranges from minutes to years and may be increased or decreased depending on cell needs.

Type of protein	Half life ($T_{1/2}$)
Muscle proteins	160 days
Body proteins	80 days
Serum proteins	10 days
Liver proteins	6 days
LDH, cytochromes	>100 hours
HMG-CoA reductase	<2 hours
Oncogene product	<2 hours

Signals for protein breakdown

- ❖ It is not clear what triggers or initiates protein degradation.
- ❖ However, some **structural features** of proteins which serve as signals for their degradation have been identified.

1. Ubiquitination

- Ubiquitin is a small protein present in eukaryotes.
- Ubiquitin attachment to a protein serve as signal for degradation of that protein.

2. PEST sequences

- PEST sequence **proline (P)**, **glutamate (E)**, **serine (S)** and **threonine (T)** rich region of a protein serves as mark for the degradation of the protein.

3. N-terminal amino acids

- Presence of amino acids like **arginine** or **lysine**, **phenyl alanine** and **aspartate** at amino terminal serve as signal for degradation of the protein.

4. Oxidized amino acids

Enzymes of protein degradation

1. Protease and peptidases

- Several intracellular proteases and peptidases are involved in protein degradation. Proteases like **trypsin**, **chymase**, **elastase**, **collagenase** and **calpains** hydrolyzes proteins forming oligo peptides.
- **Peptidases** further degrade oligo peptides to dipeptides.
- Finally aminoacids are formed by the action of **dipeptidases** on dipeptides.

2. Cathepsins

- They are hydrolytic enzymes present in lysosomes. They are responsible for the degradation of glycoprotein hormones and other intracellular proteins.

3. Megapain

- It is a high molecular mass protease present in liver and skeletal muscle. It degrades proteins in presence of ubiquitin.

4. Multicatalytic proteases (Proteasome)

- They pocess multicatalytic function and are present in the cytosol of most of mammalian cells.

Amino acid pool

- **Amino acid pool** is the amount of free amino acids distributed throughout the body.
- Blood plasma contains most of the 20 amino acids, however the concentrations of individual amino acids in blood are different (e.g. There are very high con'c of glutamate and alanine and very low con'c of aspartate)
- Amino acids in plasma are rapidly taken up by tissues, therefore the average half life of a plasma amino acid is about 5 minutes.

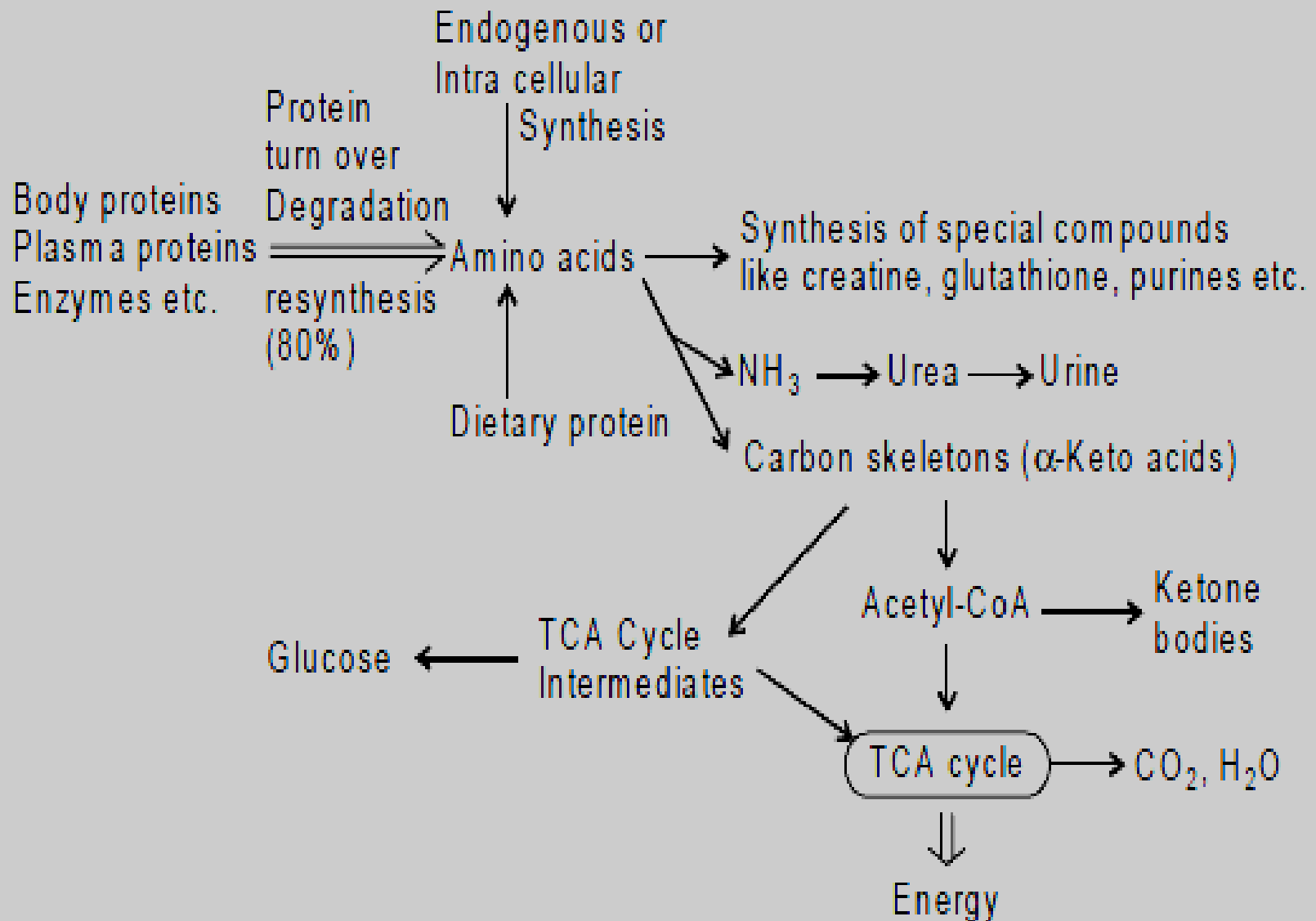
Sources of amino acid pool

1. Dietary proteins.
2. Breakdown of tissue proteins.
3. Biosynthesis of non-essential amino acids.

Fate of Amino Acids

1. Biosynthesis of structural proteins e.g tissue proteins
2. Biosynthesis of functional proteins e.g Haemoglobin , myoglobin, protein hormones and enzymes
3. Biosynthesis of small peptides of biological importance e.g glutathione, endorphins and enkephalins.
4. Biosynthesis of non protein nitrogenous compounds (NPN) such as urea, uric acid, creatine, creatinine, nucleotides and ammonia.
5. Catabolism to ammonia and α -keto acids.

Sources and metabolic fates of amino acids



Amino acid catabolism

- In the adult human body, only 20% of amino acids formed from protein degradation are catabolised to generate energy.
- The remaining 80% amino acids are used for protein biosynthesis.
- Excessive amino acids consumed from diet are also used for energy production because amino acids can not be stored in the body.
- Amino acid catabolism occurs in two main stages.
 1. Removal of amino group of amino acids as ammonia which is converted to urea and excreted in urine.
 2. Conversion of the carbon skeletons of amino acids into intermediates of TCA cycle and acetyl-CoA.

Metabolic Circumstances of Amino Acid Catabolism

- In animals, amino acids undergo oxidative degradation in three different metabolic circumstances:
 1. *During the normal synthesis and degradation of cellular proteins (**protein turn over**)*
 2. *In high protein diet, the ingested amino acids exceed the body's needs for protein synthesis, the surplus is catabolized (**amino acids cannot be stored**).*
 3. *During starvation or in uncontrolled diabetes mellitus, when carbohydrates are either unavailable or not properly utilized, cellular proteins are used as fuel.*
- Amino acids are transaminated to form α -keto acids, (carbon skeletons).
- The α -keto acids undergo oxidation to CO_2 and H_2O to provide energy for brain, skeletal muscle, and other tissues or, provide three- and four-carbon units for gluconeogenesis.

Metabolic Fates of Amino Groups

- In the cytosol of hepatocytes, amino groups from most amino acids are transferred to α -ketoglutarate to form glutamate, which enters mitochondria and gives up its amino group to form NH_4^+
- Excess ammonia generated in other tissues is converted to the amide nitrogen of glutamine, which enters the mitochondria $\rightarrow$ urea cycle.
- In skeletal muscle, excess amino groups are generally transferred to pyruvate to form alanine.

Deamination of amino acids

- This involves the first stage of amino acid catabolism.
 - There are several ways for the removal of α -amino group of amino acids.
1. Transamination followed by oxidative deamination (**transdeamination**).
 2. Oxidative deamination.
 3. Non-oxidative deamination.

1. Transamination followed by oxidative deamination (**transdeamination**)

a) **Transamination**

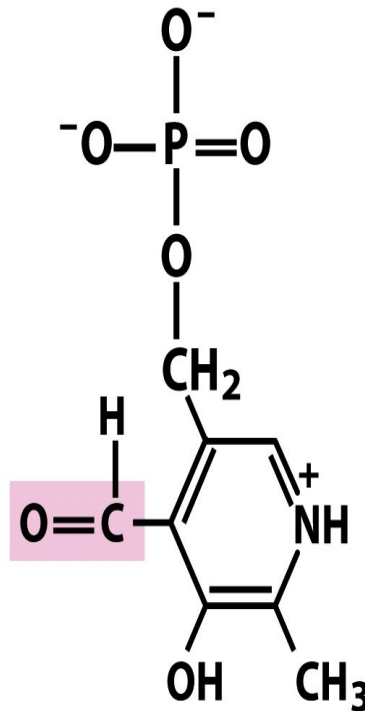
- This is the first step in the catabolism of most of the amino acids.
- The enzymes involved are known as **transaminases**, which are present in mitochondria and cytosol of most tissues and contain **pyridoxal phosphate** as prosthetic group.
- They transfer α -amino group to an acceptor mostly to **α -keto glutarate**.
- They are named according to substrate whose aminogroup is transferred to α -keto glutarate.(e.g. aspartate transaminase).
- Though there are several transaminases, most important among them are **aspartate amino transferase** and **alanine amino transferase**.

Note: Glutamate serves as a collecting point of α -amino groups.

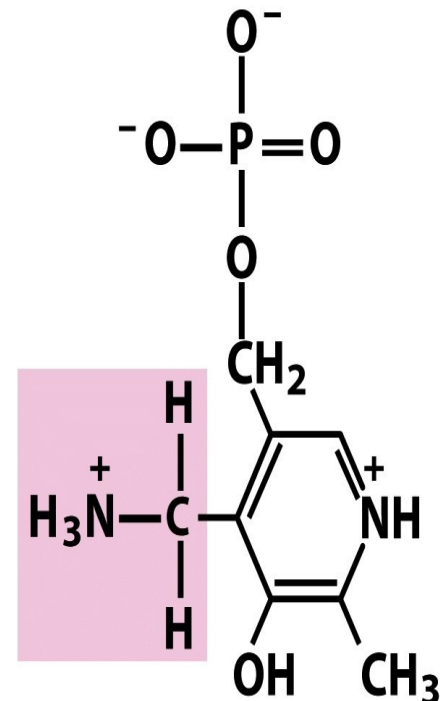
- ◆ Glutamate then functions as an amino group donor for **biosynthetic pathways** or for **excretion pathways** that lead to the elimination of nitrogenous waste products (**e.g. Urea Cycle**).

Structure of Pyridoxal Phosphate and Pyridoxamine Phosphate

- Intermediate, enzyme-bound carrier of amino groups.
- Aldehyde form can react reversibly with amino groups.
- Aminated form can react reversibly with carbonyl groups.



**Pyridoxal phosphate
(PLP)**



**Pyridoxamine
phosphate**

b) Oxidative deamination

- ❖ In hepatocytes, glutamate is transported from the cytosol into mitochondria, where it undergoes **oxidative deamination** catalyzed by **L-glutamate Dehydrogenase**.
- ❖ The combined action of an **aminotransferase** and **glutamate dehydrogenase** is referred to as **transdeamination**.
- ❖ **Glutamate dehydrogenase** is influenced by an array of allosteric modulators (e.g. ADP is a positive modulator while GTP is a negative modulator).
- ❖ Mutations that alter the allosteric binding site for GTP lead to a human genetic disorder called **hyperinsulinism-hyperammonemia syndrome**, characterized by elevated levels of ammonia in the bloodstream and hypoglycemia.

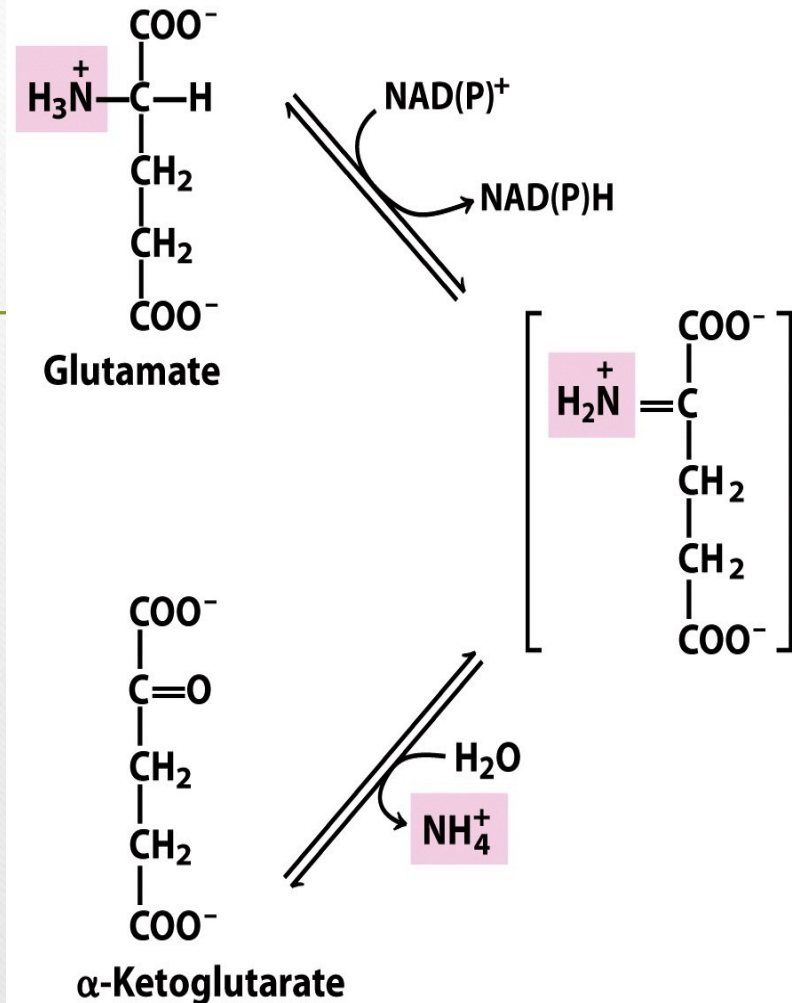


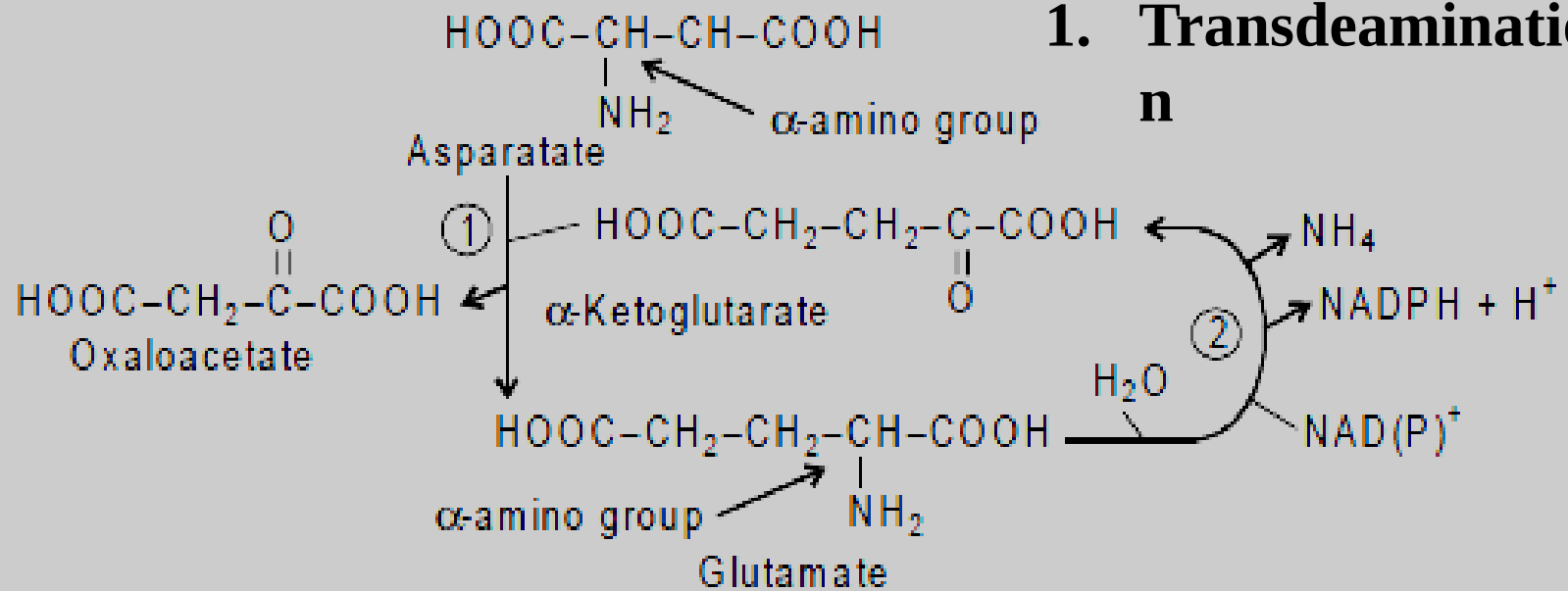
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2. Oxidative deamination

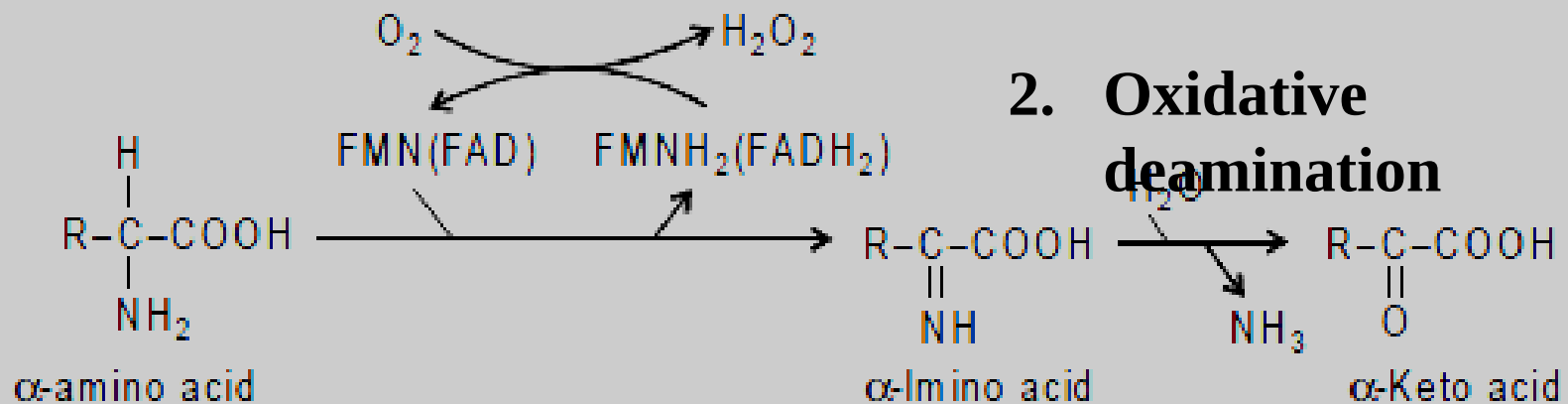
- This is an NAD^+ independent oxidative deamination.
- This type of oxidative deamination is catalyzed by **amino acid oxidases**, Present in the liver and kidney.
- They are of two types: **D-amino acid oxidase** and **L-amino acid oxidase**, which depend on FAD and FMN respectively.
- The reaction mechanism of these oxidases involves first oxidation of aminoacid to **iminoacid**, followed by hydrolytic loss of ammonia to produce α -keto acid.

Note: The activity of **L-amino acid oxidase** is low, therefore its contribution to ammonia production minimal.

1. Transdeamination



2. Oxidative deamination

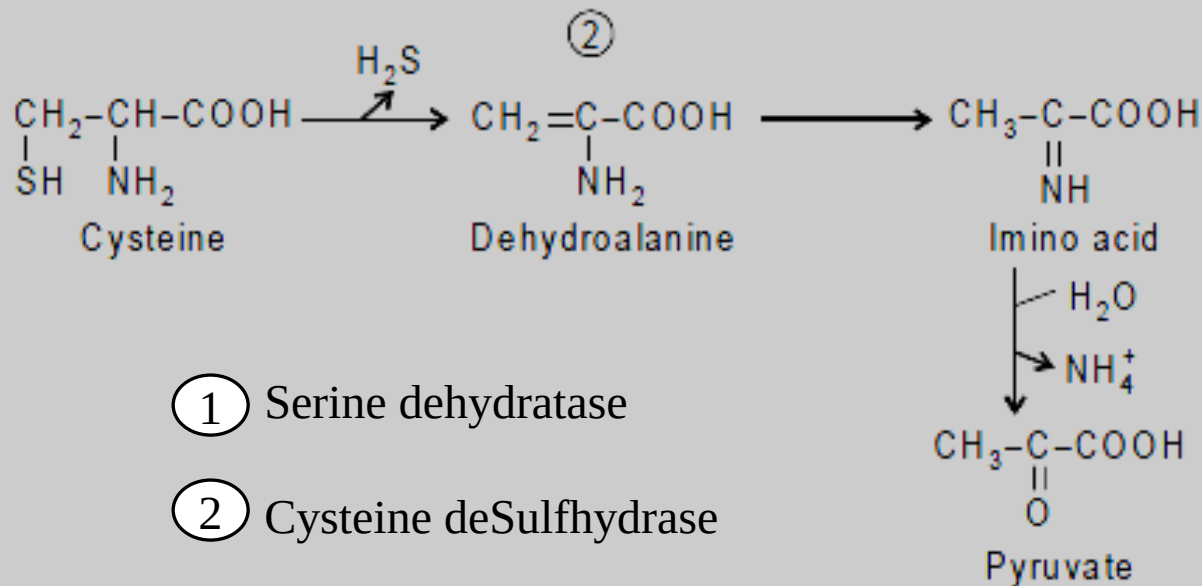
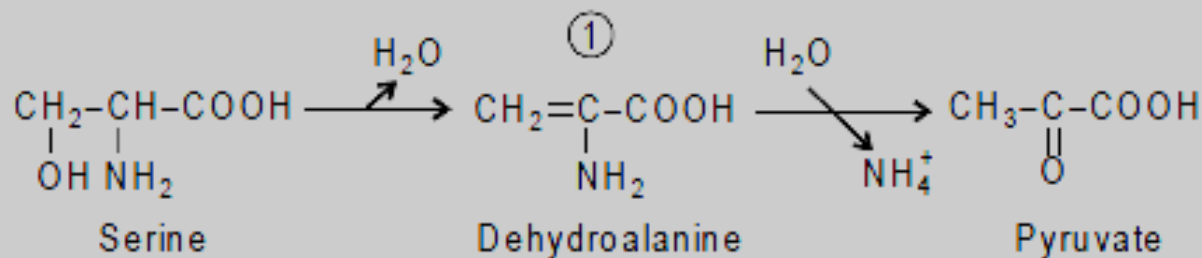


(1) Aspartate amino transferase

(2) Glutamate dehydrogenase

3. Non-oxidative deamination

- Non-oxidative deamination of amino acids is catalyzed by specific enzymes.

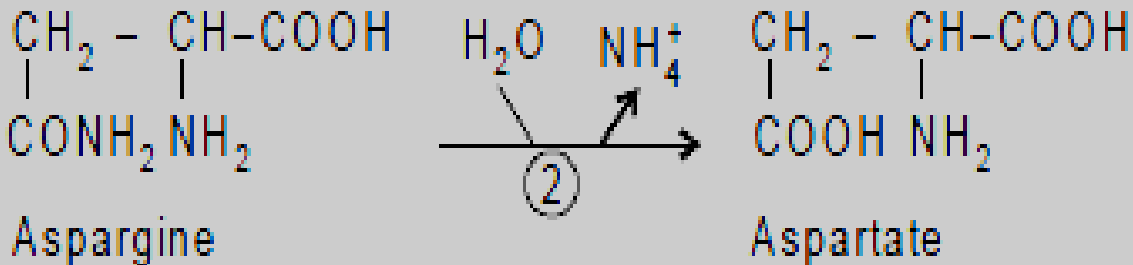
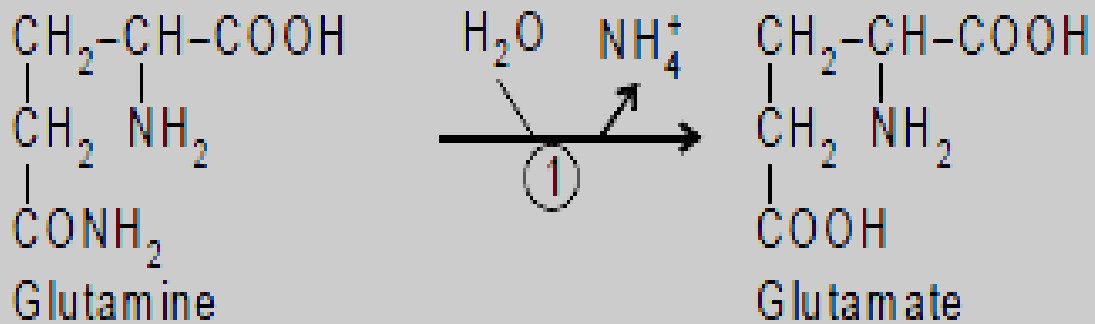


① Serine dehydratase

② Cysteine deSulfhydrase

Removal of amide nitrogen of glutamine and asparagine

- Specific deamidinases removes amide nitrogen of glutamine and asparagine as ammonia.
- They are **glutaminase** and **asparaginase**, present in the kidney.

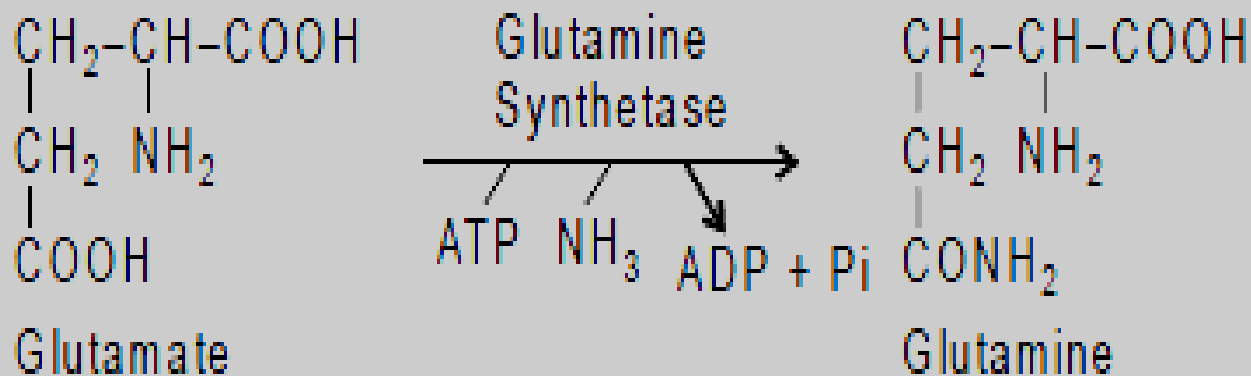


Transport of Ammonia

- Ammonia produced in the tissues is transported in the form of glutamine and alanine, because free ammonia is toxic even in trace amounts.

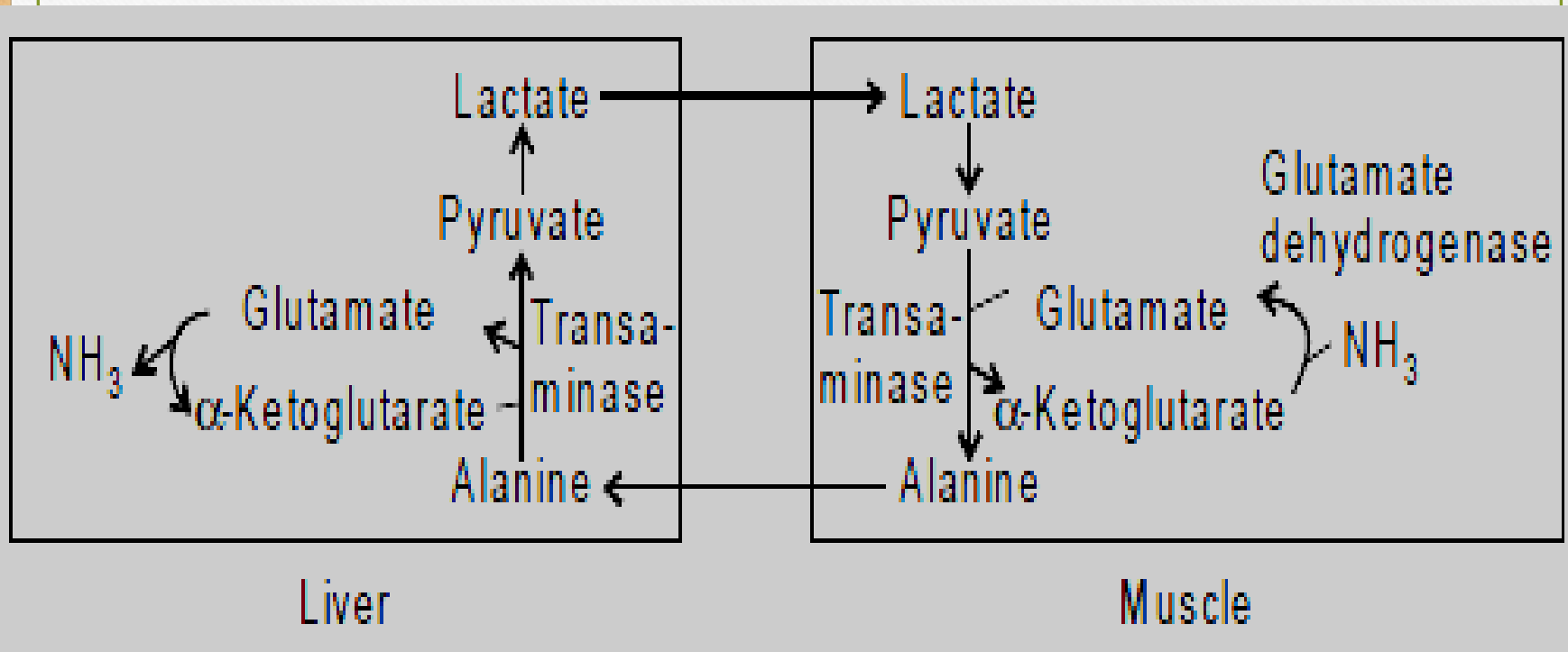
Transport of ammonia as glutamine

- From the brain and other peripheral organs except **muscle** ammonia released from amino acids is transported to liver and kidney as amide nitrogen of glutamine.
- The enzyme **glutamine synthetase** catalyzes the formation of glutamine from glutamate and ammonia in an ATP dependent reaction.



Transport of ammonia as alanine

- Since **glutamine synthetase** activity is low in skeletal muscle ammonia released from deamination of amino acids is transported in the form alanine to liver.



Amino group catabolism.

- Under normal conditions liver maintains blood ammonia level within limits by rapidly converting it into **urea**.
- Thus ammonia produced from glutamine by the action of **glutaminase** and **deamination** of amino acids is converted to urea in the **UREA CYCLE**.

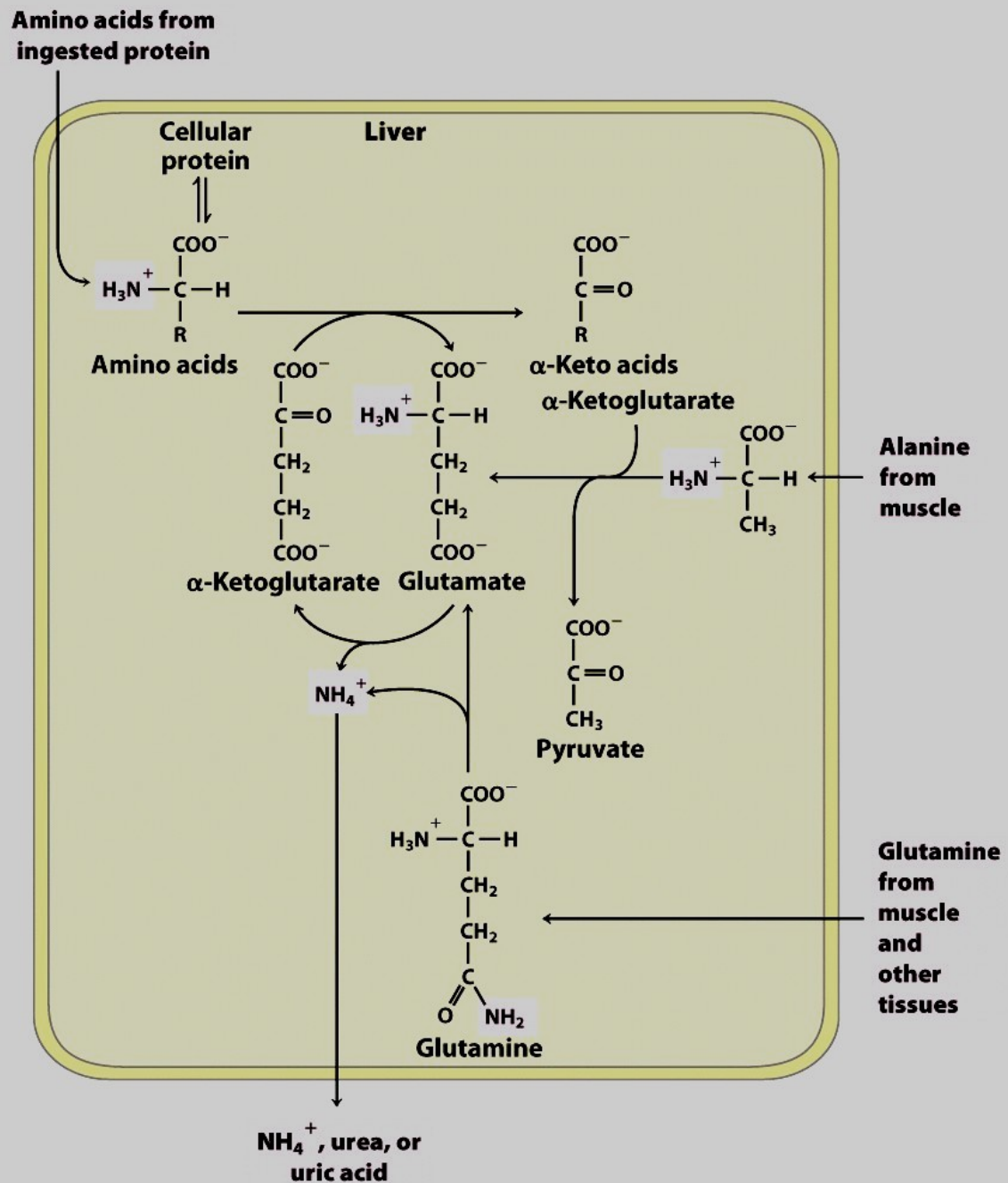


Figure 18-2a

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Urea cycle

- The site for urea synthesis is the liver.
- The reactions leading to formation of urea from ammonia were proposed by **Krebs** and **Henseleit**. Hence, urea cycle is also called as **Krebs-Henseleit cycle**.
- Formation of urea from ammonia in urea cycle occurs in five reactions. However the first reaction is not a part of the cycle but for the continuation of the cycle, the first reaction is essential.
- Synthesis of urea from ammonia is an energy dependent process.
- Enzymes of urea cycle are present in mitochondria and cytosol.
- First two reactions of urea formation occurs in mitochondria while the remaining reactions occur in cytosol.

Reaction Sequence Of Urea Cycle

Formation of carbamoyl phosphate

- First reaction leading to urea formation is condensation of **ammonia** and HCO_3^- at the expense of two high-energy bonds to form **carbamoyl phosphate**.
- The reaction is catalyzed by a mitochondrial **carbamoyl phosphate synthetase-I**.
- The enzyme requires N-acetyl glutamate, which is synthesized from acetyl-CoA and glutamate in the liver and Mg^{2+} .
- 2ATP molecules are hydrolyzed to 2 ADP and 2Pi in the first reaction and of the 2Pi one Pi is consumed in the reaction.
- Since the product of the reaction **carbamoyl phosphate** is high energy compound its formation thermodynamically pulls subsequent reactions of urea cycle towards urea formation.

Reactions of urea cycle

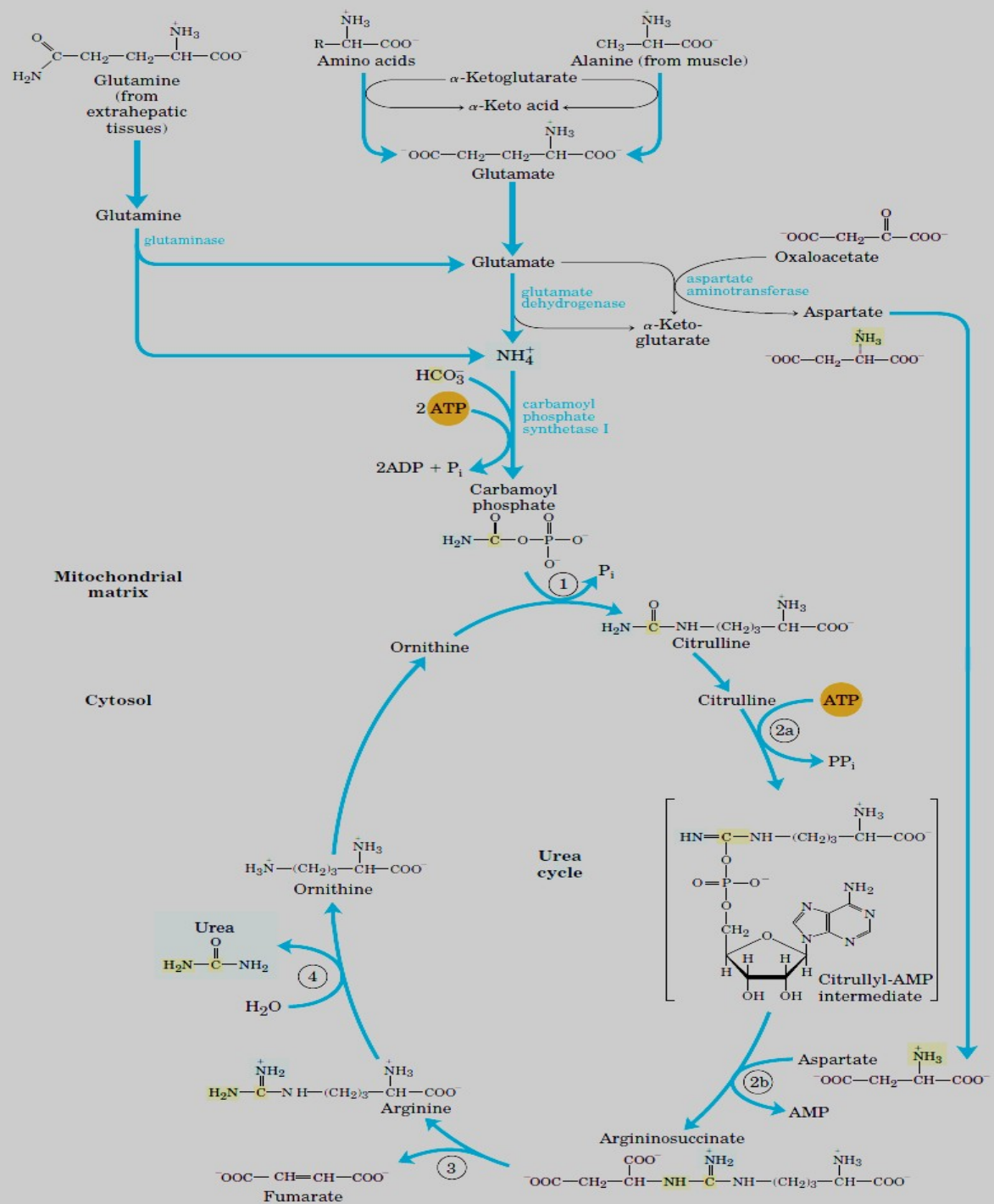
- The first reaction of urea cycle is catalyzed by **ornithine transcarbamoylase**. It condenses **carbamoyl phosphate** and **ornithine** to form **citrulline** (this occurs in the mitochondria).

Note: Citrulline is then transported through specific transporters to the cytosol.

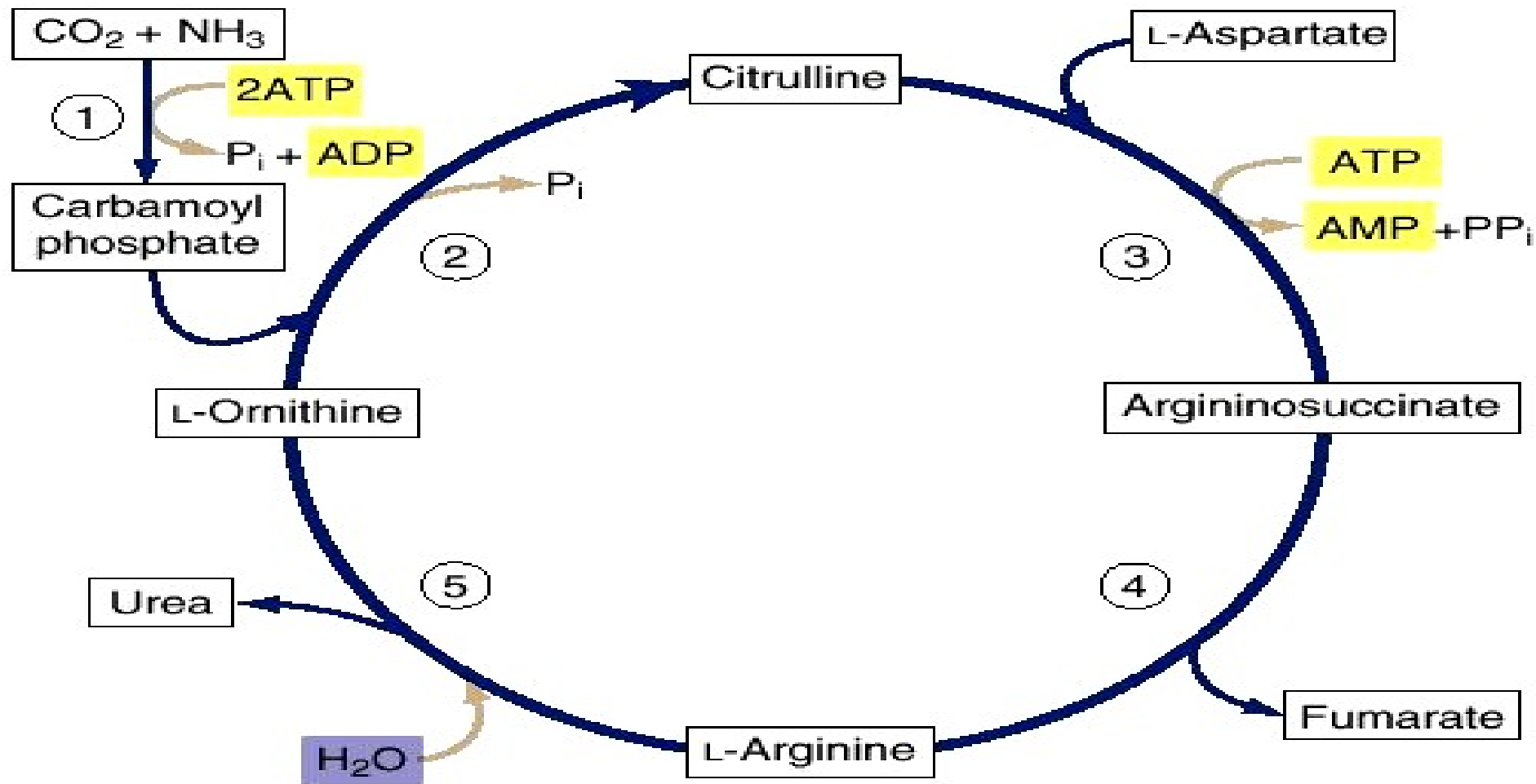
- **Arginino succinate synthetase** present in cytosol catalyzes the condensation of citrulline and aspartate at the expense of two high energy bonds to form **argininosuccinate**.
- In the third reaction of urea cycle **argininosuccinate** is cleaved by **argininosuccinase** to **arginine** and **fumarate**.
- Regeneration of **ornithine** and formation of **urea** from **arginine** is the final reaction of urea cycle. This reaction is catalyzed by **arginase**.



Nitrogen Excretion and the Urea Cycle



Urea cycle



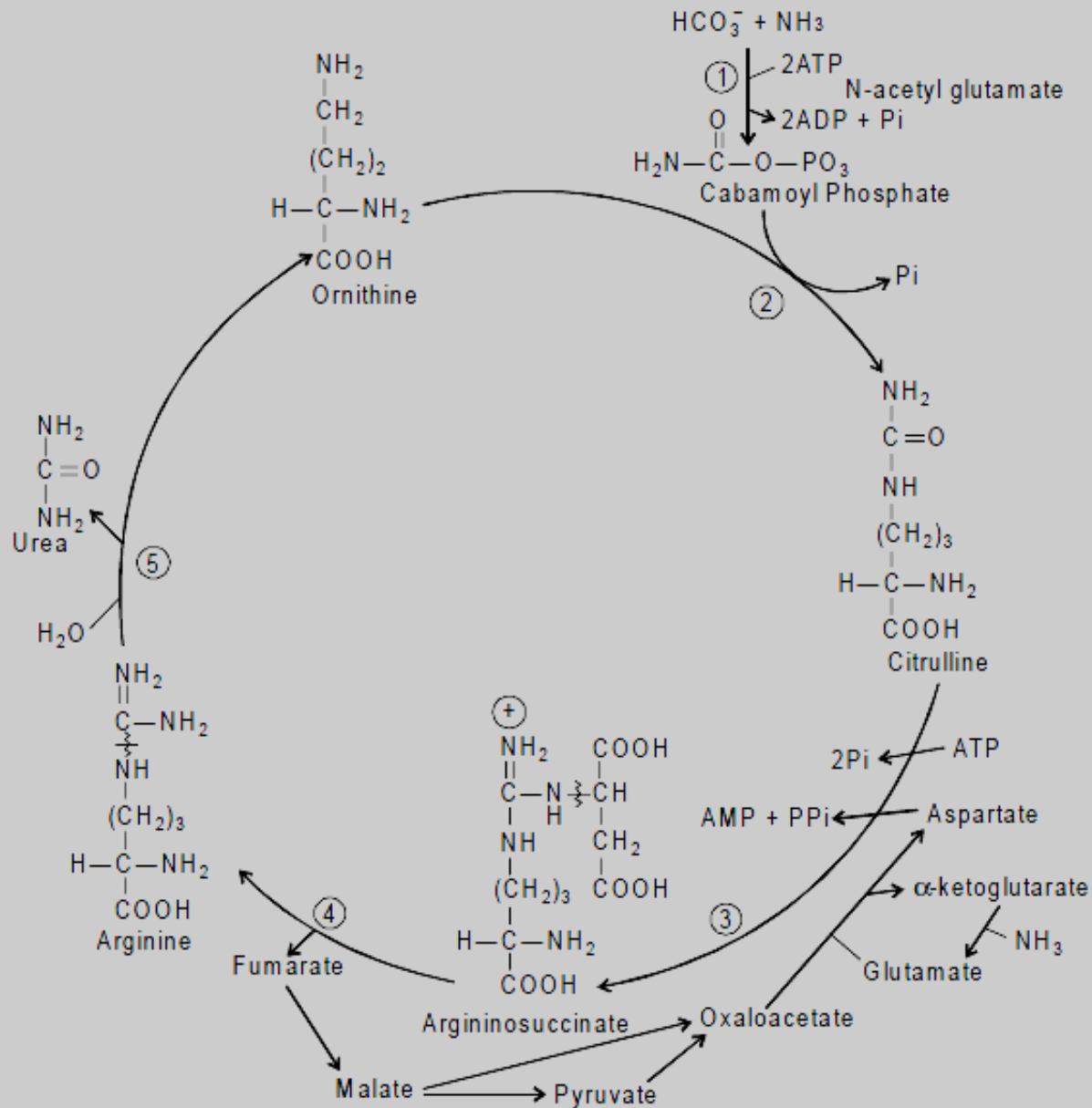
KEY TO ENZYMES (Circled Numbers)

1. Carbamoyl-phosphate synthase (ammonia)
2. Ornithine carbamoyltransferase
3. Argininosuccinate synthase
4. Argininosuccinate lyase
5. Arginase

The Reactions in the Urea Cycle

- **①** ornithine + carbamoyl phosphate $\Rightarrow$ citrulline
(entry of the first amino group).
citrulline passes into the cytosol.
- **2a** citrulline + ATP $\Rightarrow$ citrullyl-AMP + PPi
- **2b** citrullyl-AMP + Aspartate $\Rightarrow$ argininosuccinate + AMP
(entry of the second amino group).
- **3** argininosuccinate $\Rightarrow$ arginine + fumarate
 - fumarate enters the citric acid cycle.
- **4** arginine $\Rightarrow$ urea + ornithine
 - Ornithine passes to the mitochondria to continue the cycle

Fate of fumarate from the urea cycle



Fumarate is recycled in urea cycle by converting it to aspartate.

Aspartate – Argininosuccinate Shunt Links Urea Cycle and Citric Acid Cycle

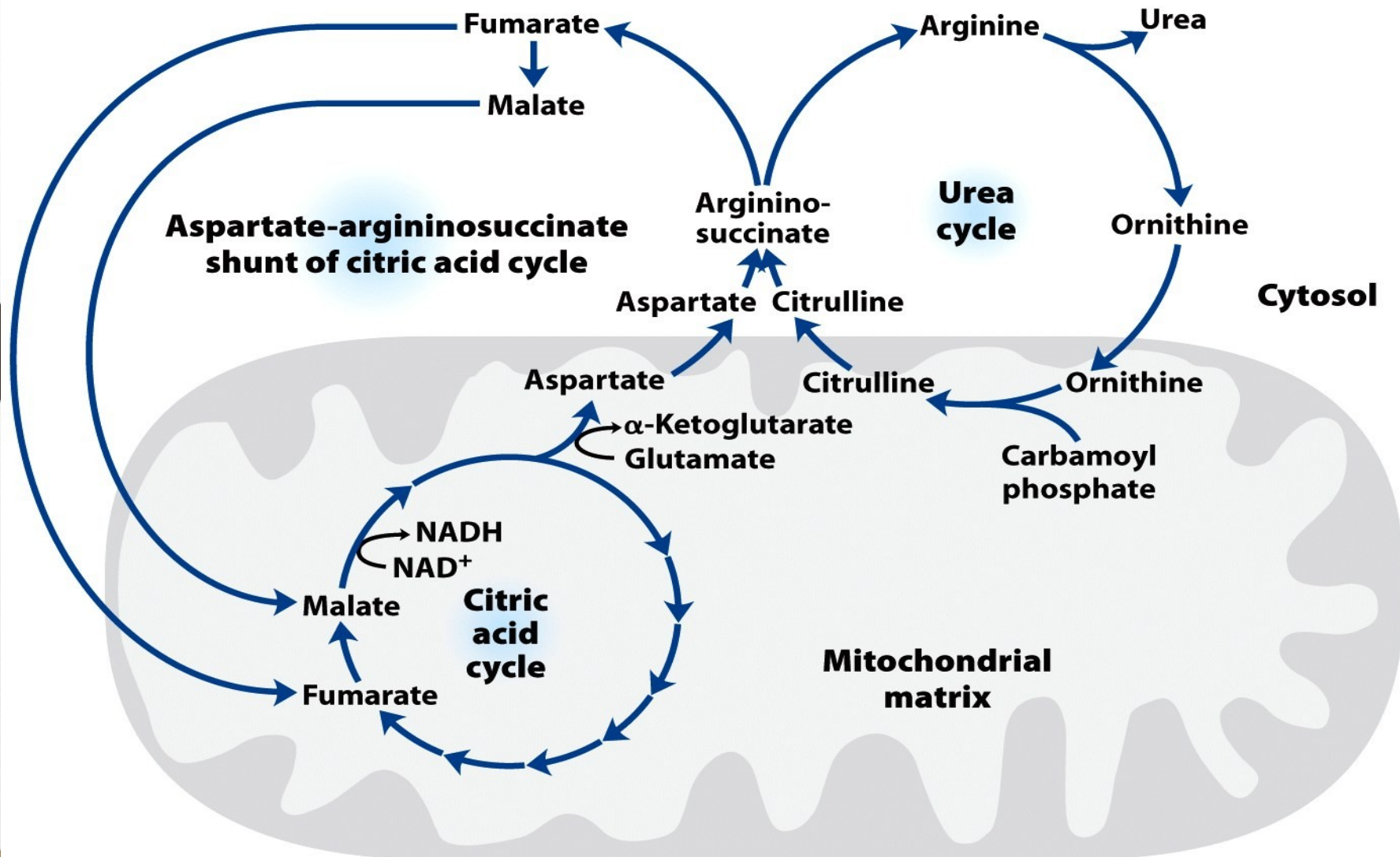


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Regulation of urea formation

- Formation of urea is regulated by activity of **carbamoyl phosphate synthetase-I**. This enzyme catalyzes committed step in urea synthesis.
- N-acetylglutamate is an allosteric activator of the enzyme.
- High protein in take leads to high levels of N-acetylglutamate formation. Thus high protein in take influences urea formation.
- In starvation the rate of urea synthesis is also high due to increased protein breakdown.

Diseases of the urea cycle

- Urea formation is impaired in several inherited diseases due to deficiency of enzymes of urea cycle.
- Since the urea cycle converts ammonia to urea these disorders may cause ammonia intoxication.
- Some common clinical symptoms seen in these diseases are vomiting, irritability, lethargy, seizures, mental retardation, coma and early death.

1. **Hyper ammonemia Type I**

- It is due to deficiency of enzyme **carbamoyl phosphate synthetase-I**.
- Mental retardation is the main symptom of this condition.

2. **Hyper ammonemia Type II**

- It is most common among others. It is due to deficiency of enzyme **ornithine trans carbamoylase**.
- In this condition carbamoyl phosphate accumulates and diverted to pyrimidine formation resulting to excretion of orotic acid and uracil in urine.
- Glutamate also accumulates in this condition.

3. **Citrullinemia**

- This condition is due to the absence of enzyme **argininosuccinate synthetase**. Hence citrulline accumulates in blood and excreted in urine.

4. **Argininosuccinic aciduria**

- **Argininosuccinase** is absent in this condition. So, **argininosuccinate** accumulates in blood and excreted in urine.

5. **Hyper argininemia**

- This condition is due to low **arginase** activity. Hence, arginine accumulates and excreted in urine. However some urea may be excreted in urine due to kidney

Treatment

- Treatment of urea cycle disorders involves removal of excess ammonia from blood and reduction of dietary nitrogen load.
- Peritoneal dialysis is employed to clear ammonia from blood.
- Administration of compounds which can increase nitrogen excretion is another line of treatment.
- Benzoic acid and phenyl acetate are two such compounds used in the treatment.

Fates of Nitrogen in Organisms

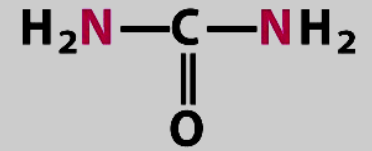
- Plants conserve almost all the nitrogen
- Many aquatic vertebrates release ammonia to their environment through;
 - passive diffusion from epithelial cells
 - active transport via gills
- Many terrestrial vertebrates and sharks excrete nitrogen in the form of urea
 - Urea is far less toxic than ammonia
 - Urea has very high solubility
- Some animals, such as birds and reptiles excrete nitrogen as uric acid
 - Uric acid is rather insoluble
 - Excretion as paste allows to conserve water

Excretory Forms of Nitrogen



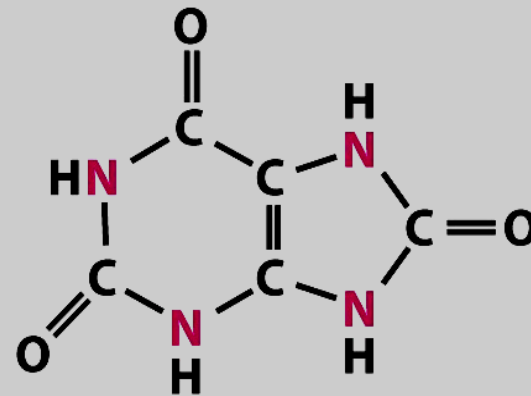
**Ammonia (as
ammonium ion)**

**Ammonotelic animals:
most aquatic vertebrates,
such as bony fishes and
the larvae of amphibia**



Urea

**Ureotelic animals:
many terrestrial
vertebrates; also sharks**



Uric acid

**Uricotelic animals:
birds, reptiles**

Figure 18-2b

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Overview of Amino Acid Catabolism

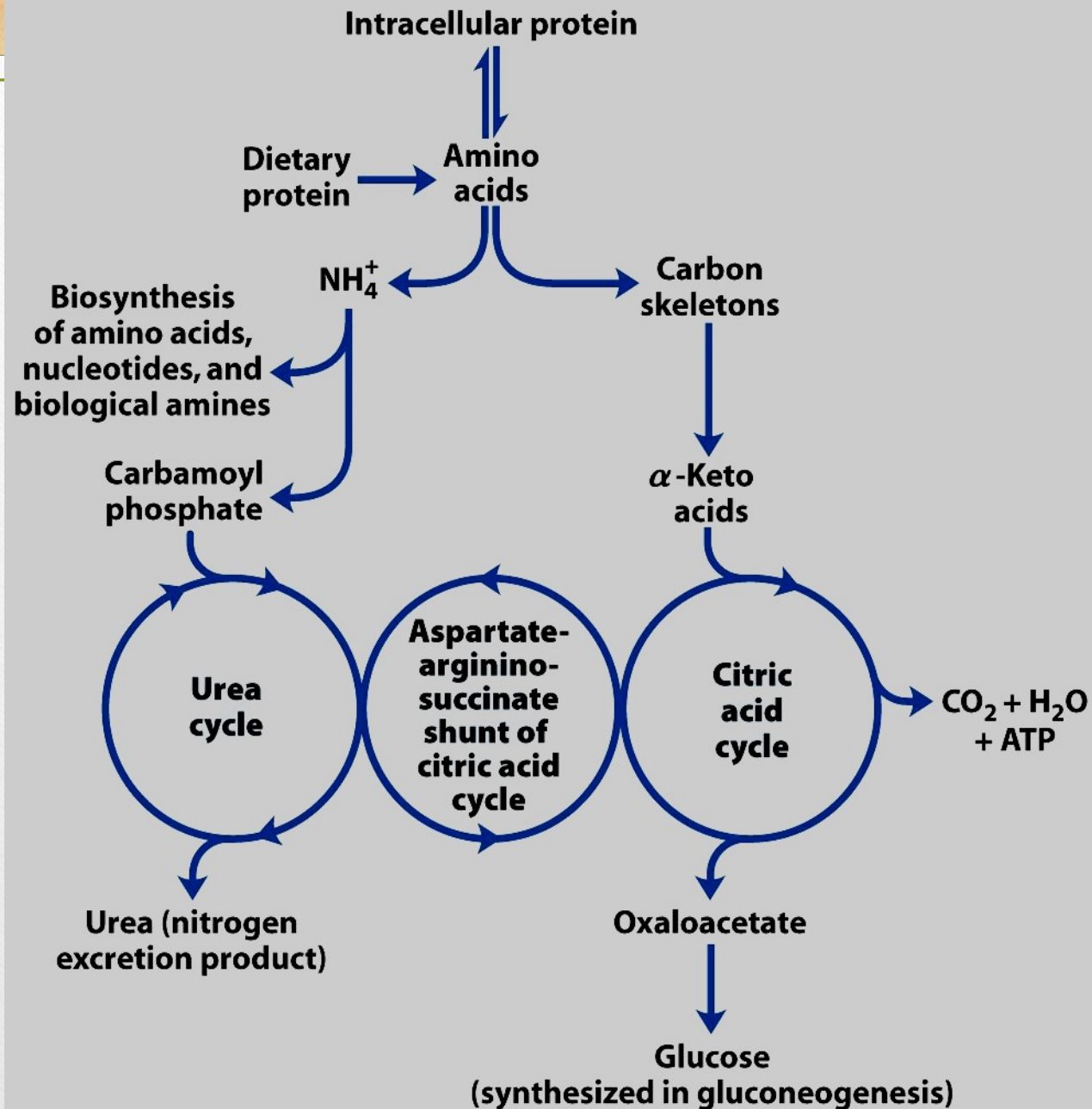


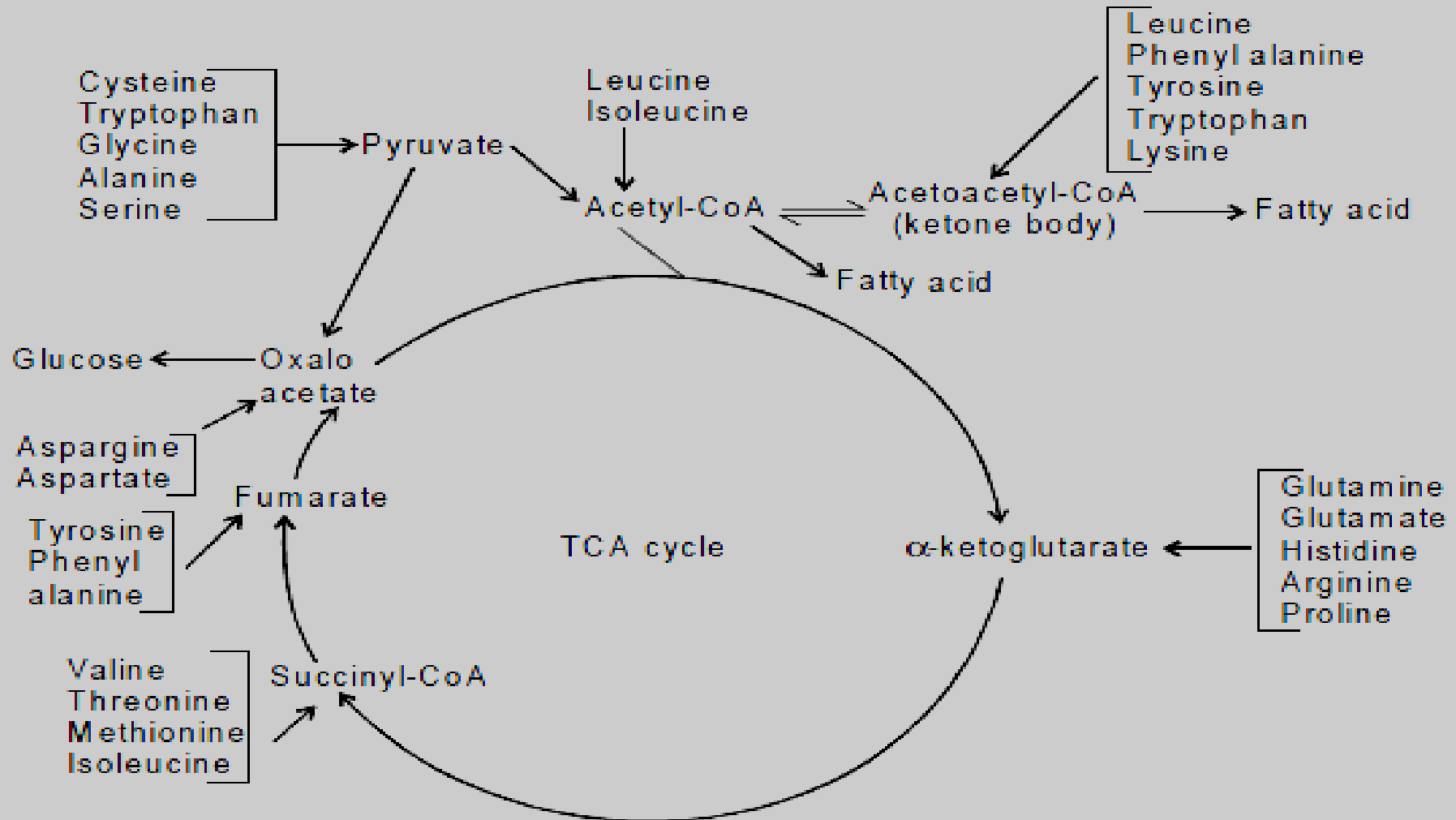
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Metabolic fates of carbon skeletons of Amino acids

- ✓ Initial deamination of amino acids produces carbon skeletons of amino acids.
- ✓ The carbon skeletons of twenty amino acids are converted to seven compounds.
- ✓ These seven compounds are ultimately used for the formation of carbohydrates or fat like substances.



- Amino acids also are classified based on the metabolic fate of their carbon skeletons.

1. **Glucogenic amino acids**

- The carbon skeletons of these amino acids are converted to either glucose or intermediates of TCA cycle.
- The products may be pyruvate, oxaloacetate, α -ketoglutarate, succinate and fumarate.
- Glycine, alanine, valine, serine, threonine, cysteine, methionine, aspartate (asparagine), glutamate (glutamine), histidine, arginine and proline are glucogenic amino acids.
- Note: all non-essential amino acids are glucogenic.

2. **Ketogenic amino acids**

- The carbon skeletons of these amino acids give rise to fat like substances or intermediates of fatty acid catabolism.
- The products may be aceto acetyl-CoA and acetyl-CoA. Leucine is the only ketogenic amino acid.
- Isoleucine, phenylalanine, tyrosine, tryptophan and lysine are also ketogenic amino acids.

3. **Glucogenic and ketogenic amino acids**

- The carbon skeletons of these amino acids are converted to glucose or intermediates of TCA cycle and fat like substance.
- The products may be pyruvate, succinate, fumarate and acetyl-CoA.
- Isoleucine, phenylalanine, tyrosine, tryptophan and lysine are glucogenic and ketogenic

Fate of Individual Amino Acids

- Seven to **acetyl-CoA**
 - Leu, Ile, Thr, Lys, Phe, Tyr, Trp
- Six to **pyruvate**
 - Ala, Cys, Gly, Ser, Thr, Trp
- Five to **α -ketoglutarate**
 - Arg, Glu, Gln, His, Pro
- Four to **succinyl-CoA**
 - Ile, Met, Thr, Val
- Two to **fumarate**
 - Phe, Tyr
- Two to **oxaloacetate**
 - Asp, Asn

Six Amino Acids Are Degraded to Pyruvate

- The carbon skeletons of six amino acids are converted in whole or in part to pyruvate.
- The pyruvate can then be converted to either acetyl-CoA (a ketone body precursor) or oxaloacetate (a precursor for gluconeogenesis). Thus amino acids catabolized to pyruvate are both ketogenic and glucogenic.
- The six are **alanine, tryptophan, cysteine, serine, glycine, and threonine.**

6 Amino Acids -> Pyruvate, Ala, Gly, Ser, Cys, Trp, Thr.

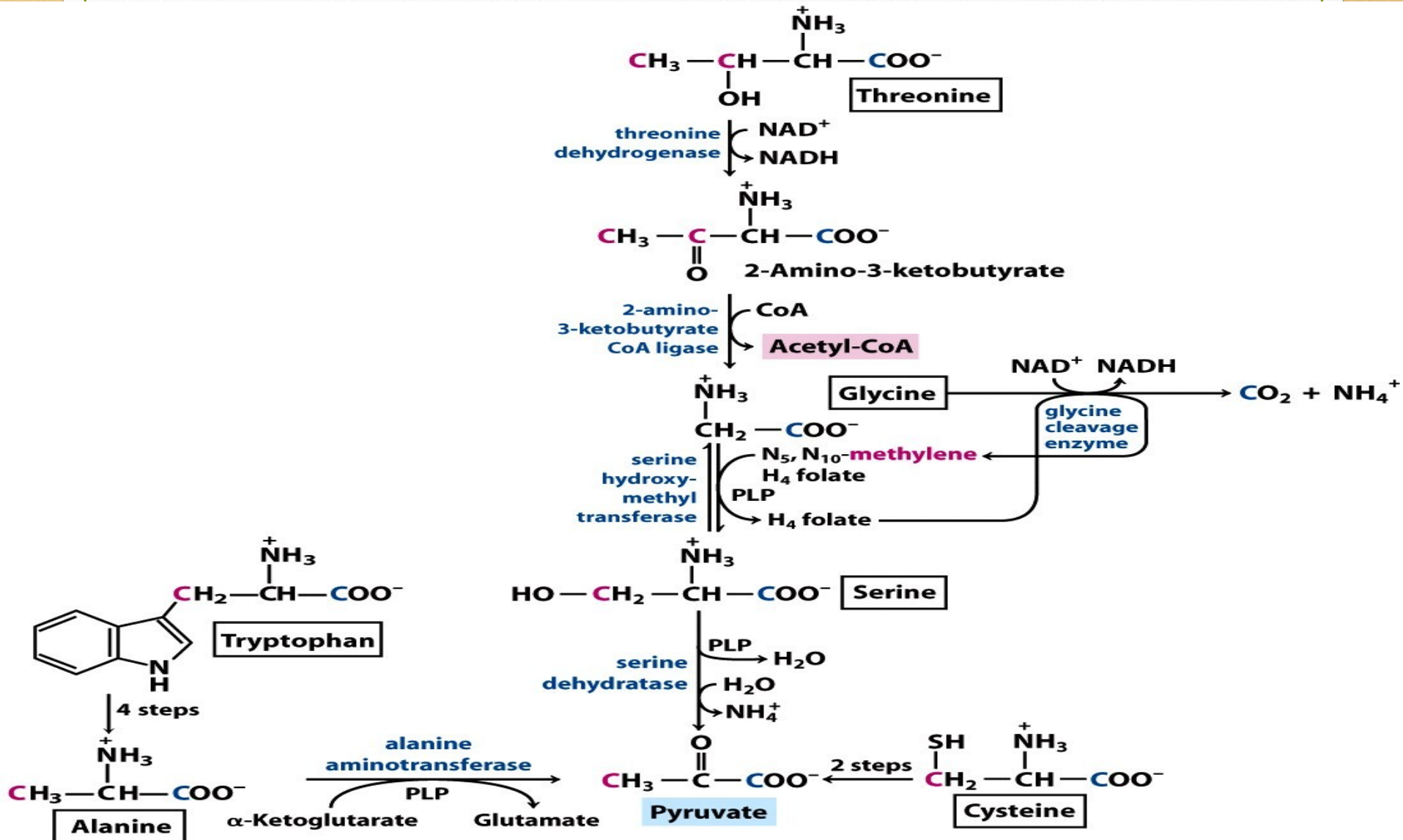


Figure 18-19

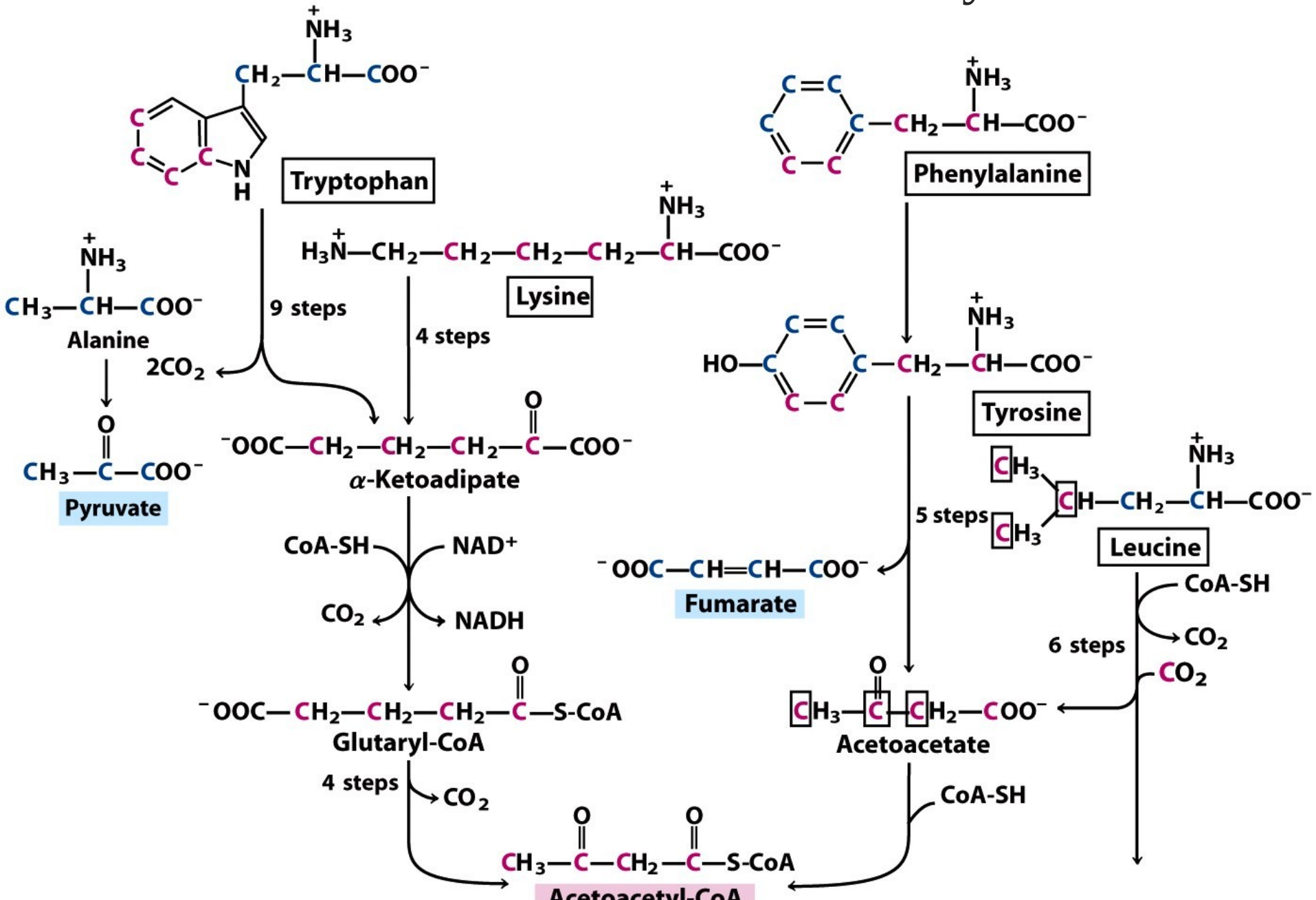
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Seven Amino Acids Are Degraded to Acetyl-CoA

- Portions of the carbon skeletons of seven amino acids—**tryptophan, lysine, phenylalanine, tyrosine, leucine, isoleucine, and threonine**—yield acetyl-CoA and/or acetoacetyl-CoA, the latter being converted to acetyl-CoA.
- Some of the intermediates in tryptophan catabolism are precursors for the synthesis of other biomolecules, including **nicotinate**, a precursor of **NAD** and **NADP** in animals; **serotonin**, (a neurotransmitter in vertebrates) and **indoleacetate** (a growth factor in plants).
- The breakdown of phenylalanine is **noteworthy** because genetic defects in the enzymes of this pathway lead to several inheritable human diseases.
- After its hydroxylation to tyrosine, phenylalanine is also the precursor of **dopamine (a neurotransmitter)** and of **norepinephrine** and **epinephrine** (hormones secreted by the adrenal medulla).
- Melanin, the black pigment of skin and hair, is also derived from tyrosine.

7 AAs -> Acetyl CoA



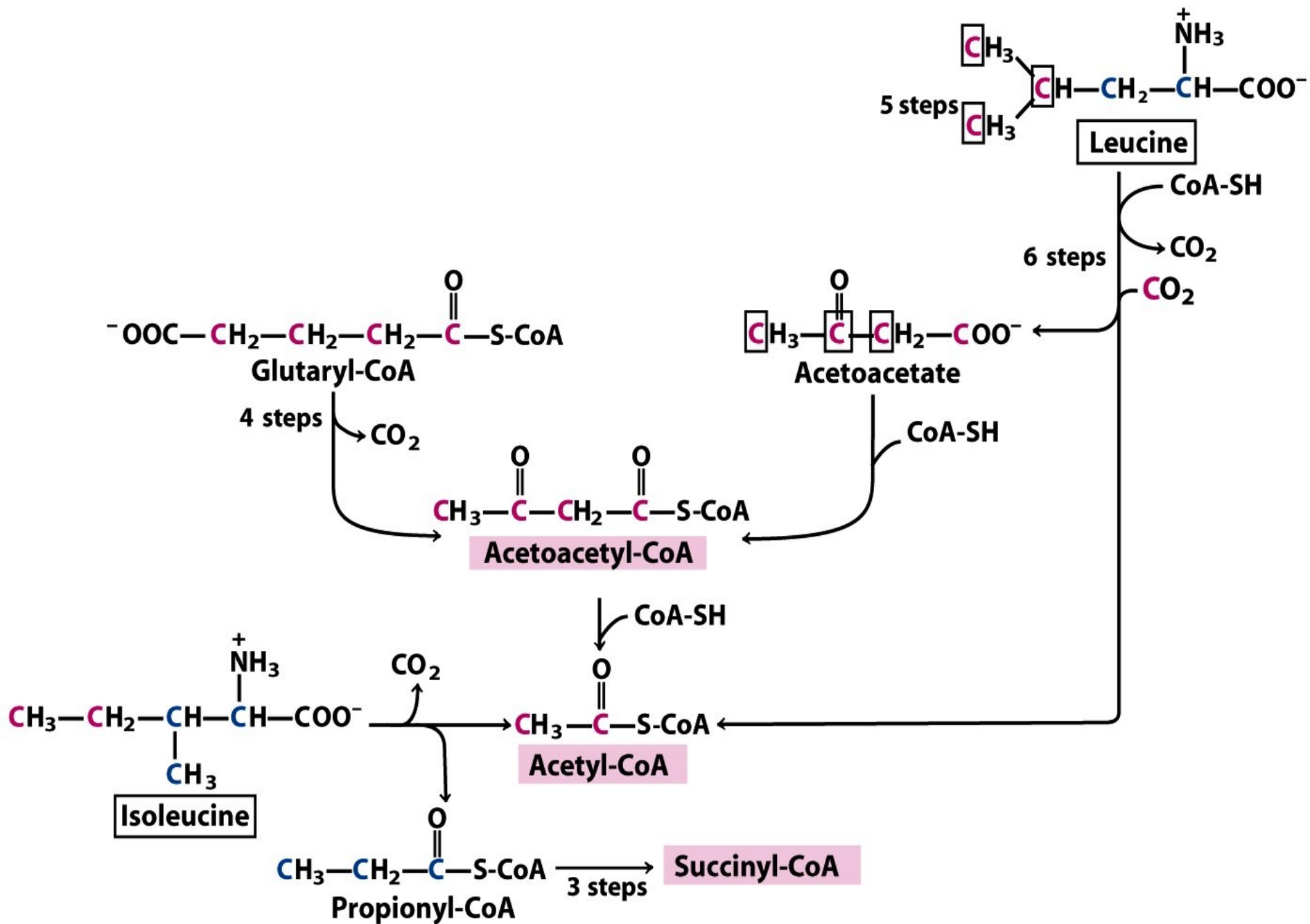
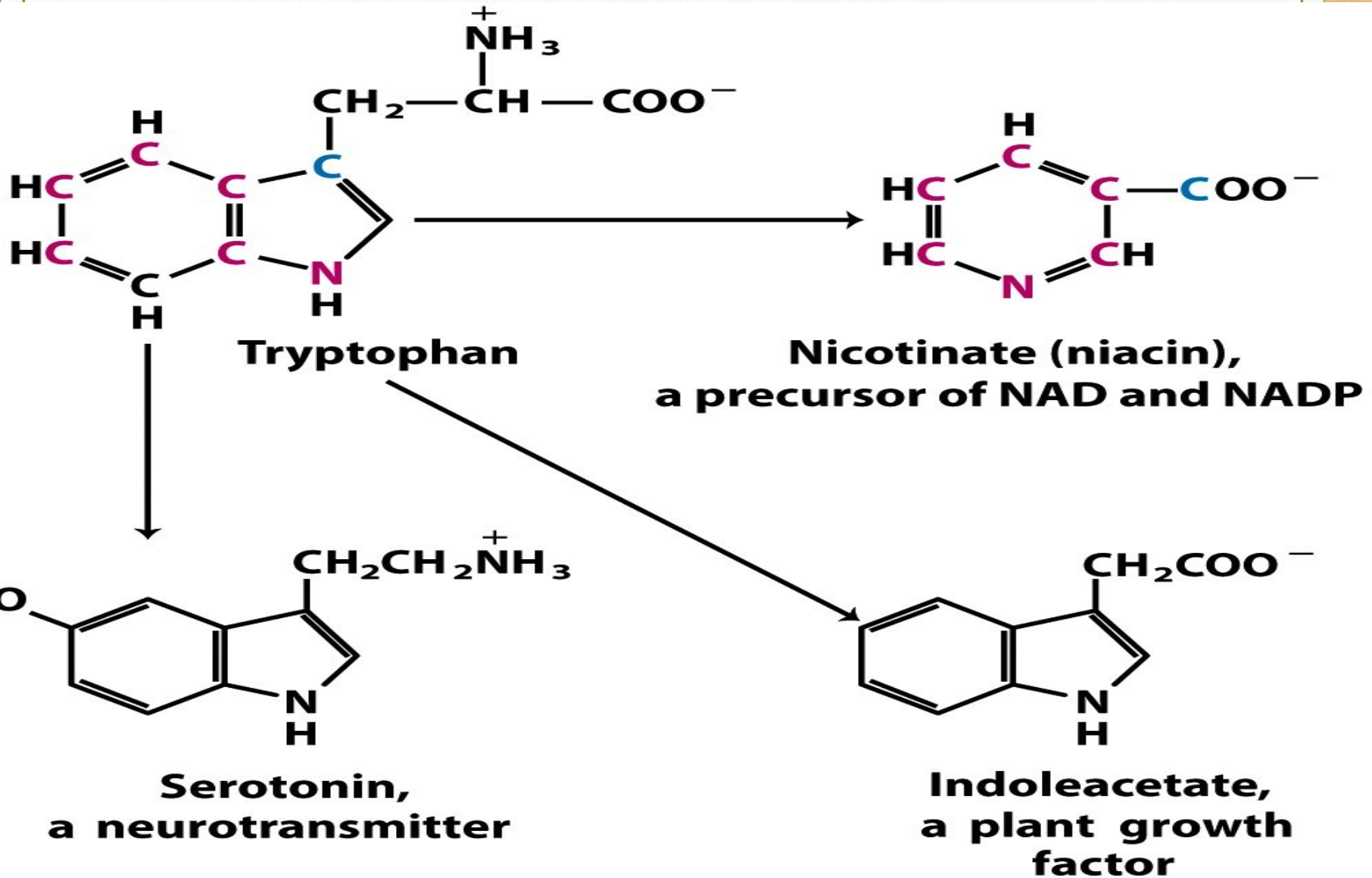


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Indole Ring Recycling



Metabolic Diseases – Defects of Aromatic AA degradation

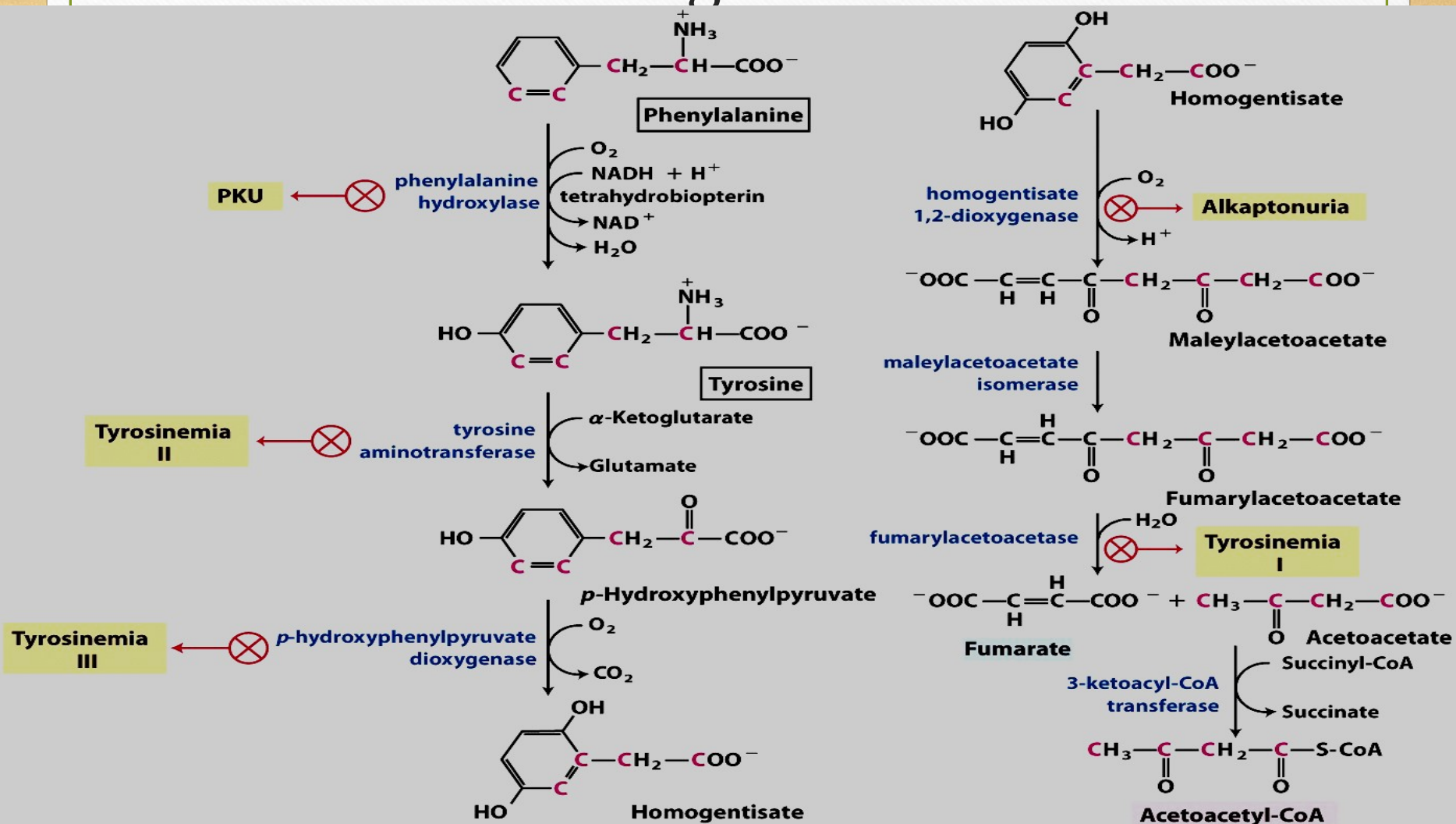


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Amino acids $\rightarrow$ α -ketoglutarate

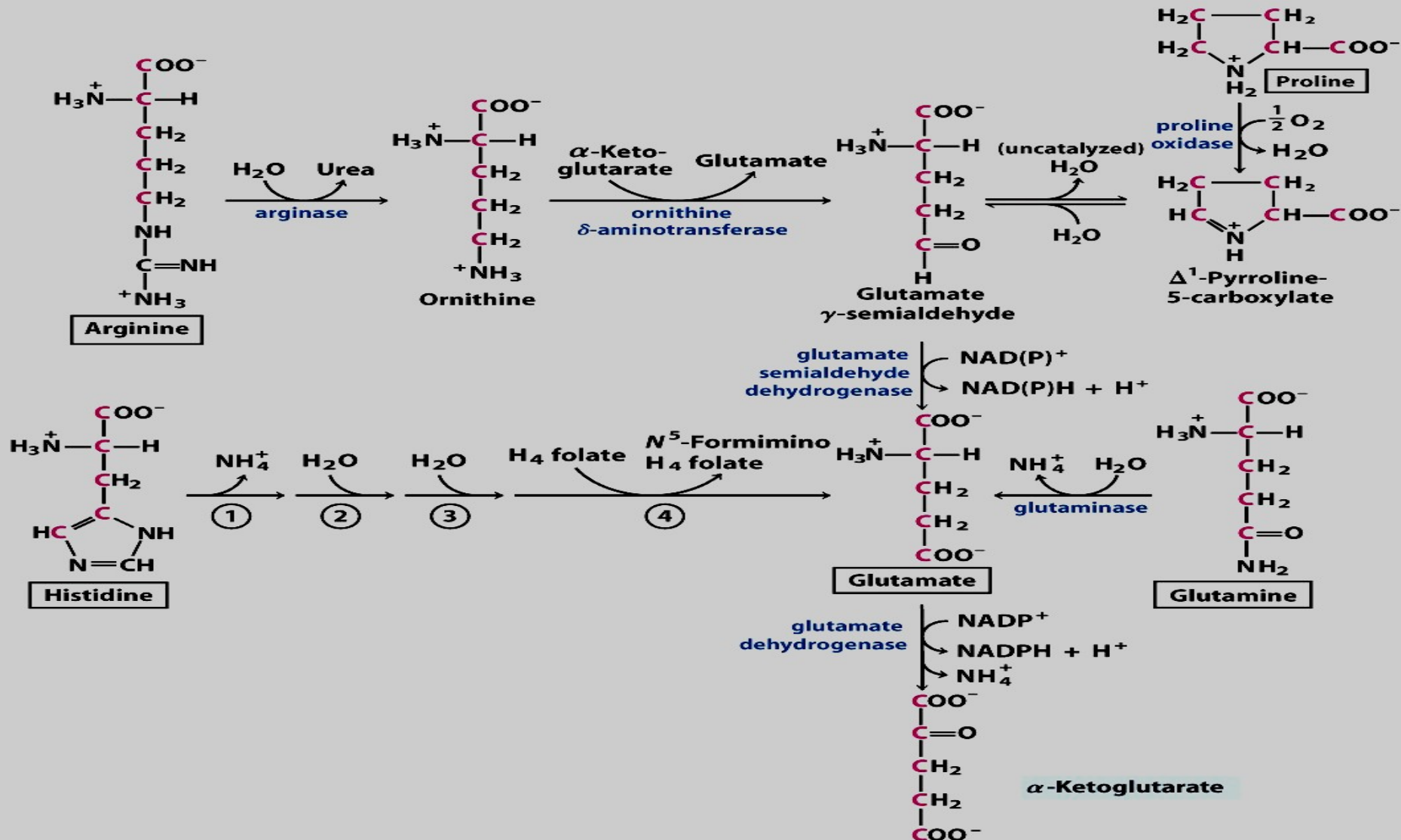


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Amino acids → Succinyl-CoA

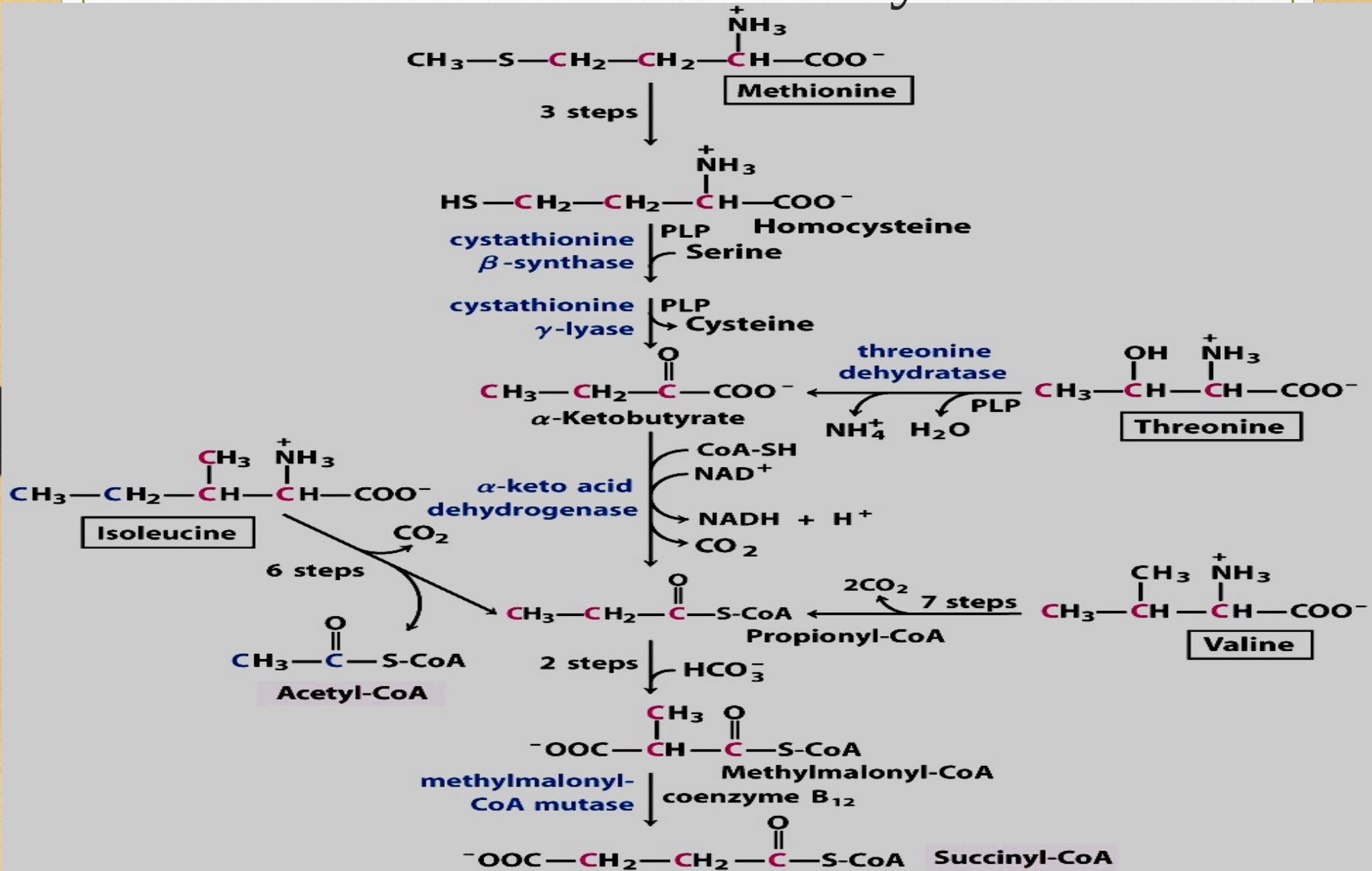


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Asp and Asn to oxaloacetate.

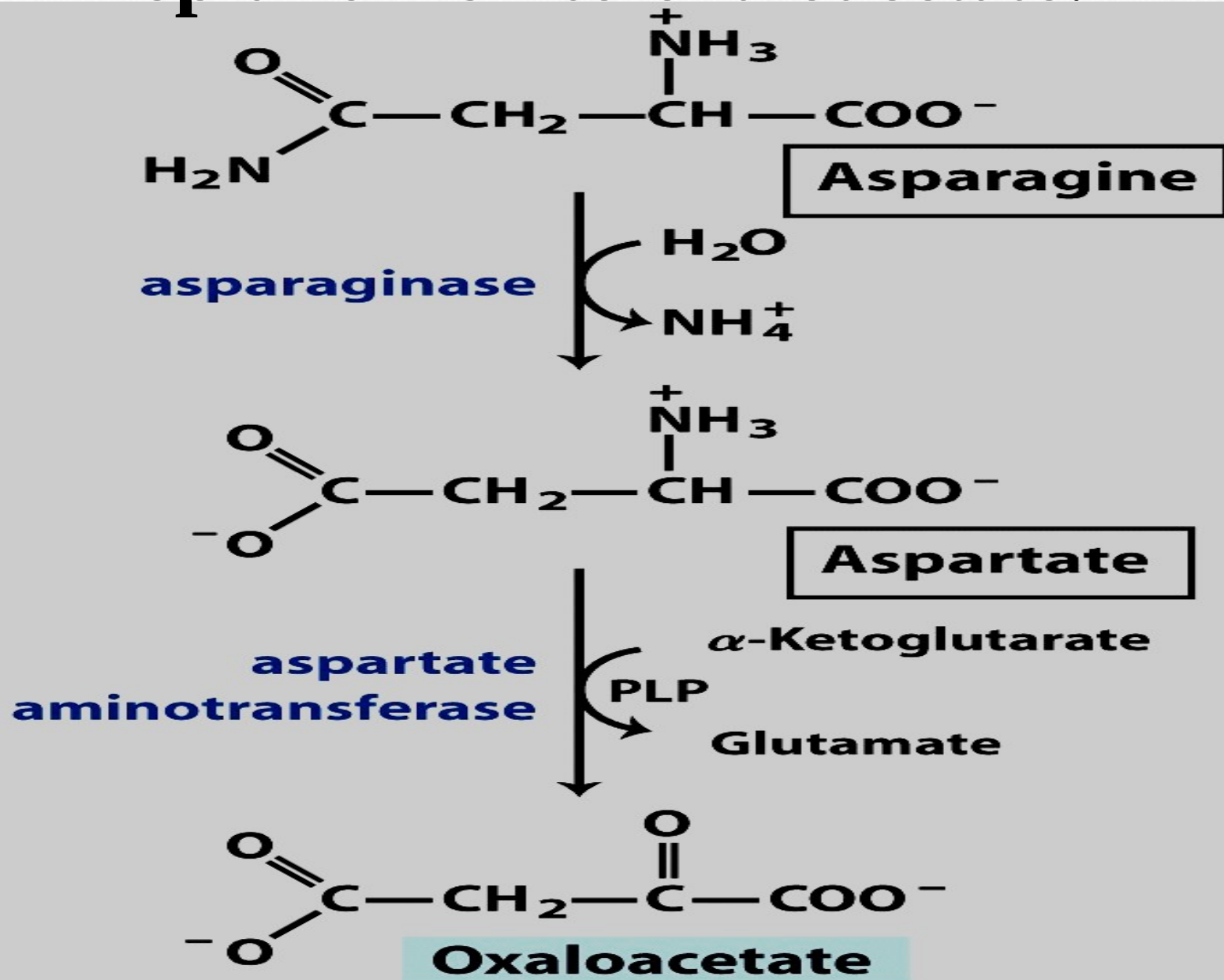


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Summary of Amino Acid Catabolism

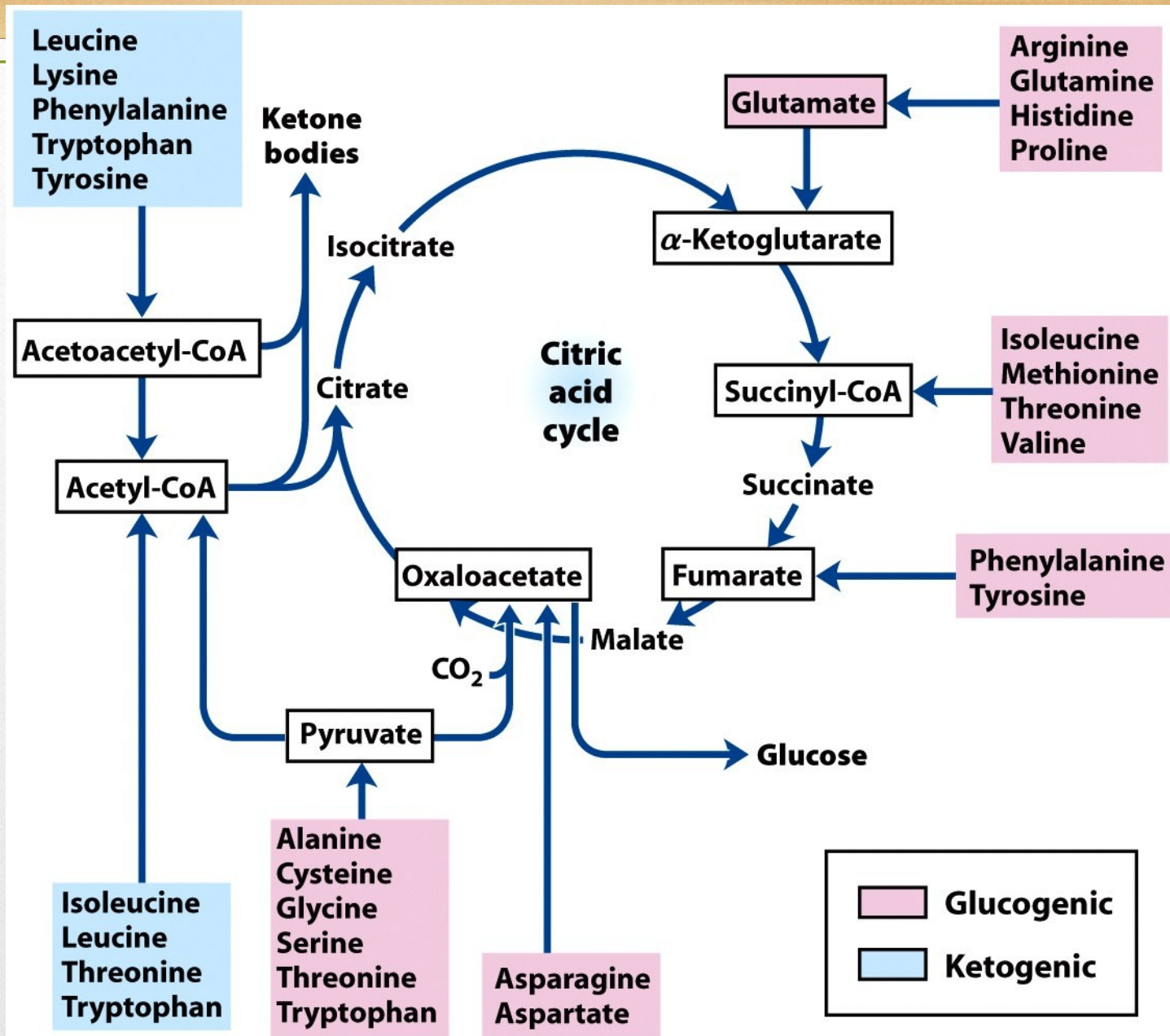


Figure 18-15

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